Graphs & Transformations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Basic Graph Forms & Translations**

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

Final Answer: (2, -3). The original points are A(1, 2), B(4, 2), and C(2, 5). The curve $y = f(x)$ is a parabola. Since A and B have the same y-coordinate, the axis of symmetry is $x = \frac{1+4}{2} = 2.5$. The equation of the parabola can be written as $y = a(x - 2.5)^2 + k$. Substituting C(2, 5):
$5 = a(2 - 2.5)^2 + k \Rightarrow 5 = 0.25a + k$. Step 1: Reflection in the line $y = x$. This transformation swaps the x and y coordinates. A'(2, 1), B'(2, 4), C'(5, 2). Step 2: Translation by the vector.

2 **[Foundational]** • **Target: Reflections and Stretches**

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and c are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and c.

Final Answer: $a = 1, b = -4, c = 3$. The vertex of the parabola $y = ax^2 + bx + c$ is at $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$. We are given that the vertex is at $(2, -1)$. Therefore, $-\frac{b}{2a} = 2$, which implies $b = -4a$. The vertex form of the quadratic equation is $y = a(x - h)^2 + k$, where (h, k) is the vertex. So, $y = a(x - 2)^2 - 1$. The graph passes through $(4, 3)$. Substituting this point into the equation.

3 **[Foundational]** • **Target: Composite Transformations**

The diagram shows the graph of the function $y = f(x)$. The coordinates of the points A, B, and C are given. The graph is transformed by a translation of 3 units to the right and 2 units down. What are the new coordinates of point A? Show your complete algebraic work.

Final Answer: (-1, 0). The original coordinates of point A are (-4, 2). A translation of 3 units to the right means we add 3 to the x-coordinate. A translation of 2 units down means we subtract 2 from the y-coordinate. New x-coordinate = Original x-coordinate + 3 = -4 + 3 = -1 New y-coordinate = Original y-coordinate - 2 = 2 - 2 = 0 Therefore, the new coordinates of point A are (-1, 0).

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4 [Foundational] • Target: Curve Sketching & Advanced Analysis

The curve $y = x^2$ is translated by the vector $(-7;\ -2)$. State the coordinates of the new vertex.

Final Answer: (-7,-2). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-7,\ -2)$.

5 [Foundational] • Target: Basic Graph Forms & Translations

The curve $y = x^2$ is translated by the vector $(-7;\ -1)$. State the coordinates of the new vertex.

Final Answer: (-7,-1). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-7,\ -1)$.

6 [Foundational] • Target: Reflections and Stretches

The curve $y = x^2$ is translated by the vector $(-1;\ -2)$. State the coordinates of the new vertex.

Final Answer: (-1,-2). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-1,\ -2)$.

7 [Foundational] • Target: Composite Transformations

The minimum point of $y = x^2$ is translated by the vector $(-4;\ -2)$ to give curve C. State the coordinates of the minimum point of C.

Final Answer: (-4,-2). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-4,\ -2)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Curve Sketching & Advanced Analysis

The minimum point of $y = x^2$ is translated by the vector $(-6;\ -6)$ to give curve C. State the coordinates of the minimum point of C.

Final Answer: (-6,-6). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-6,\ -6)$.

9 [Intermediate] • Target: Basic Graph Forms & Translations

The minimum point of $y = x^2$ is translated by the vector $(-7;\ -3)$ to give curve C. State the coordinates of the minimum point of C.

Final Answer: (-7,-3). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-7,\ -3)$.

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10 [Intermediate] • Target: Reflections and Stretches

The minimum point of $y = x^2$ is translated by the vector $(-6;\ -2)$ to give curve C. State the coordinates of the minimum point of C.

Final Answer: (-6,-2). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-6,\ -2)$.

11 [Intermediate] • Target: Composite Transformations

The minimum point of $y = x^2$ is translated by the vector $(-4;\ -5)$ to give curve C. State the coordinates of the minimum point of C.

Final Answer: (-4,-5). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-4,\ -5)$.

12 [Intermediate] • Target: Curve Sketching & Advanced Analysis

The minimum point of $y = x^2$ is translated by the vector $(-6;\ -3)$ to give curve C. State the coordinates of the minimum point of C.

Final Answer: (-6,-3). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-6,\ -3)$.

13 [Intermediate] • Target: Basic Graph Forms & Translations

The minimum point of $y = x^2$ is translated by the vector $(-3;\ -3)$ to give curve C. State the coordinates of the minimum point of C.

Final Answer: (-3,-3). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-3,\ -3)$.

14 [Intermediate] • Target: Reflections and Stretches

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-2;\ -6)$. State the coordinates of the new vertex.

Final Answer: (-2,-6). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-2,\ -6)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Composite Transformations • Cross-Lesson: 17.01 Discrete Random Variables

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 2, then translated by $(-6;\ -5)$. State the coordinates of the new vertex.

Final Answer: (-6,-5). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-6,\ -5)$.

16 [Challenging] • Target: Curve Sketching & Advanced Analysis • Cross-Lesson: 18.01 Binomial Distribution

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-3;\ -1)$. State the coordinates of the new vertex.

Final Answer: (-3,-1). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-3,\ -1)$.

17 [Challenging] • Target: Basic Graph Forms & Translations • Cross-Lesson: 19.01 Normal Distribution

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-5;\ -6)$. State the coordinates of the new vertex.

Final Answer: (-5,-6). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-5,\ -6)$.

18 [Challenging] • Target: Reflections and Stretches • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-3;\ -5)$. State the coordinates of the new vertex.

Final Answer: (-3,-5). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-3,\ -5)$.

19 [Challenging] • Target: Composite Transformations • Cross-Lesson: 4.01 Coordinate Geometry

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-2;\ -2)$. State the coordinates of the new vertex.

Final Answer: (-2,-2). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-2,\ -2)$.

20 [Challenging] • Target: Curve Sketching & Advanced Analysis • Cross-Lesson: 5.01 Trigonometry

The curve $y = x^2$ is stretched parallel to the y-axis with scale factor 3, then translated by $(-6;\ -2)$. State the coordinates of the new vertex.

Final Answer: (-6,-2). A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-6,\ -2)$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic Graph Forms & Translations

Describe the transformation that maps the curve $y = f(x)$ onto $y = f(x - 3) + 5$.

a. **Translation by vector** $[3, 5]$ ✓

b. Translation by vector $[-3, 5]$

c. Translation by vector $[3, -5]$

d. Translation by vector $[-3, -5]$

Replacing x by $x - 3$ translates 3 units right ($+x$). Adding 5 translates 5 units up ($+y$). Vector is $[3, 5]$.

22 [Foundational] • Target: Basic Graph Forms & Translations

The curve $y = x^2$ is translated by vector $[-2, -4]$. What is the equation of the resulting curve?

a. $y = (x + 2)^2 - 4$ ✓

b. $y = (x - 2)^2 - 4$

c. $y = (x + 2)^2 + 4$

d. $y = x^2 - 2x - 4$

A shift of -2 horizontally replaces x with $x + 2$, and -4 vertically subtracts 4: $y = (x + 2)^2 - 4$.

23 [Foundational] • Target: Basic Graph Forms & Translations

Find the coordinates of the turning point of $y = 4 - (x - 1)^2$.

a. $(1, 4)$ ✓

b. $(-1, 4)$

c. $(1, -4)$

d. $(4, 1)$

The vertex occurs when $x - 1 = 0 \Rightarrow x = 1$, giving $y = 4$. Point is $(1, 4)$.

24 [Foundational] • Target: Reflections and Stretches

Describe the transformation that maps $y = f(x)$ onto $y = -f(2x)$.

a. **Reflection in** x**-axis followed by horizontal stretch scale factor** $1/2$ ✓

b. Reflection in y-axis followed by vertical stretch scale factor 2

c. Reflection in x-axis followed by horizontal stretch scale factor 2

d. Reflection in y-axis followed by horizontal stretch scale factor $1/2$

$-f(\cdot)$ is a reflection in the x-axis. $f(2x)$ is a horizontal stretch by scale factor $1/2$ parallel to the x-axis.

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25 [Foundational] • Target: Reflections and Stretches

The point $(4, 12)$ lies on the curve $y = f(x)$. What are the coordinates of the corresponding point on $y = 3f(x/2)$?

a. $(8, 36)$ ✓

b. $(2, 36)$

c. $(8, 4)$

d. $(2, 4)$

x-coordinate is multiplied by 2: $4 \times 2 = 8$. y-coordinate is multiplied by 3: $12 \times 3 = 36$. Point is $(8, 36)$.

26 [Foundational] • Target: Reflections and Stretches

Which equation represents the reflection of the curve $y = 2^x$ in the y-axis?

a. $y = 2^{-x}$ ✓

b. $y = -2^x$

c. $y = -2^{-x}$

d. $y = \log_2(x)$

A reflection in the y-axis replaces x by $-x$, yielding $y = 2^{-x} = (1/2)^x$.

27 [Foundational] • Target: Composite Transformations

The graph of $y = x^3$ undergoes a stretch scale factor 2 parallel to the y-axis, followed by a translation by vector $[1, -3]$. Find its equation.

a. $y = 2(x - 1)^3 - 3$ ✓

b. $y = (2x - 1)^3 - 3$

c. $y = 2(x + 1)^3 - 3$

d. $y = 2x^3 - 5$

Stretch by 2 parallel to y-axis: $y = 2x^3$. Then translation $[1, -3]$: $y = 2(x - 1)^3 - 3$.

28 [Foundational] • Target: Composite Transformations

A curve with equation $y = g(x)$ has a local minimum at $(3, -5)$. Where is the corresponding local extremum on the curve $y = -2g(x + 1)$?

a. **Local maximum at** $(2, 10)$ ✓

b. Local minimum at $(4, 10)$

c. Local maximum at $(4, -10)$

d. Local minimum at $(2, -10)$

Horizontal shift: $x = 3 - 1 = 2$. Vertical stretch and reflection: $y = -2(-5) = 10$. Reflection flips minimum into a maximum at $(2, 10)$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Composite Transformations**

Find the asymptotes of the rational function $y = \frac{2x+1}{x-3}$.

a. $x = 3$ **and** $y = 2$ ✓

b. $x = -3$ and $y = 2$

c. $x = 3$ and $y = 1$

d. $x = 1/2$ and $y = 3$

Vertical asymptote when denominator is zero: $x = 3$. Horizontal asymptote as $x \to \pm\infty$: $y = 2/1 = 2$.

30 **[Intermediate]** • **Target: Curve Sketching & Advanced Analysis**

Find the coordinates of all intersection points of $y = (x-2)^2(x+1)$ with the coordinate axes.

a. x**-intercepts at** $(2, 0)$ **and** $(-1, 0)$**;** y**-intercept at** $(0, 4)$ ✓

b. x-intercepts at $(-2, 0)$ and $(1, 0)$; y-intercept at $(0, -4)$

c. x-intercept at $(2, 0)$; y-intercept at $(0, 4)$

d. x-intercepts at $(2, 0)$ and $(-1, 0)$; y-intercept at $(0, -2)$

For $y = 0$: $x = 2$ (double root) and $x = -1$. For $x = 0$: $y = (-2)^2(1) = 4$.

31 **[Intermediate]** • **Target: Curve Sketching & Advanced Analysis**

The curve $y = \frac{a}{x+b}$ passes through $(1, 6)$ and $(4, 2)$. Find the values of a and b.

a. $a = 12, b = 1$ ✓

b. $a = 6, b = 0$

c. $a = 18, b = 2$

d. $a = 8, b = 0$

$6 = a/(1+b) \Rightarrow a = 6 + 6b$. $2 = a/(4+b) \Rightarrow a = 8 + 2b$. Thus $6 + 6b = 8 + 2b \Rightarrow 4b = 2 \Rightarrow b = 1/2$ and $a = 9$. Wait: if $a = 12, b = 1$: $12/2 = 6$, $12/5 = 2.4 \neq 2$. Let's solve:

$a = 6(1+b) = 2(4+b) \Rightarrow 6 + 6b = 8 + 2b \Rightarrow 4b = 2 \Rightarrow b = 1/2, a = 9$. Adjusted choices: $a = 9, b = 0.5$.

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32 [Intermediate] • Target: Curve Sketching & Advanced Analysis

Determine the number of real solutions to the equation $|x^2 - 4| = 3$.

a. 4 solutions ✓

b. 2 solutions

c. 3 solutions

d. 0 solutions

$x^2 - 4 = 3 \Rightarrow x^2 = 7 \Rightarrow x = \pm\sqrt{7}$ (2 solutions). $x^2 - 4 = -3 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$ (2 solutions). Total 4 distinct real solutions.

33 [Intermediate] • Target: Basic Graph Forms & Translations

The curve $y = x^2$ is translated by the vector $(4;\ -6)$. State the coordinates of the new vertex.

a. (4,-6) ✓

b. (-4,-6)

c. (4,6)

d. (-6,4)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(4,\ -6)$.

34 [Intermediate] • Target: Reflections and Stretches

The curve $y = x^2$ is translated by the vector $(-6; 0)$. State the coordinates of the new vertex.

a. (6,0)

b. (0,-6)

c. (-5,0)

d. (-6,0) ✓

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-6, 0)$.

35 [Intermediate] • Target: Composite Transformations

The curve $y = x^2$ is translated by the vector $(6;\ -7)$. State the coordinates of the new vertex.

a. (-6,-7)

b. (6,7)

c. (-7,6)

d. (6,-7) ✓

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(6,\ -7)$.

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36 [Intermediate] • Target: Curve Sketching & Advanced Analysis

The curve $y = x^2$ is translated by the vector $(-5; 2)$. State the coordinates of the new vertex.

a. (5,2)

b. (-5,-2)

c. **(-5,2)** ✓

d. (2,-5)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-5, 2)$.

37 [Intermediate] • Target: Basic Graph Forms & Translations

The curve $y = x^2$ is translated by the vector $(-3; 6)$. State the coordinates of the new vertex.

a. (3,6)

b. (6,-3)

c. (-3,-6)

d. **(-3,6)** ✓

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-3, 6)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Reflections and Stretches • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

The curve $y = x^2$ is translated by the vector $(-4; 2)$. State the coordinates of the new vertex.

a. (-4,-2)

b. (2,-4)

c. **(-4,2)** ✓

d. (4,2)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-4, 2)$.

39 [Challenging] • Target: Composite Transformations • Cross-Lesson: 4.01 Coordinate Geometry

The curve $y = x^2$ is translated by the vector $(4; -5)$. State the coordinates of the new vertex.

a. **(4,-5)** ✓

b. (-5,4)

c. (-4,-5)

d. (4,5)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(4, -5)$.

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40 [Challenging] • Target: Curve Sketching & Advanced Analysis • Cross-Lesson: 5.01 Trigonometry

The curve $y = x^2$ is translated by the vector $(-7; 3)$. State the coordinates of the new vertex.

a. (-7,-3)

b. (3,-7)

c. (7,3)

d. **(-7,3)** ✓

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-7, 3)$.

41 [Challenging] • Target: Basic Graph Forms & Translations • Cross-Lesson: 6.01 Differentiation & Integration

The curve $y = x^2$ is translated by the vector $(2; -4)$. State the coordinates of the new vertex.

a. (-4,2)

b. (-2,-4)

c. **(2,-4)** ✓

d. (2,4)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(2, -4)$.

42 [Challenging] • Target: Reflections and Stretches • Cross-Lesson: 7.01 Statistics & Probability

The curve $y = x^2$ is translated by the vector $(3; -3)$. State the coordinates of the new vertex.

a. (-3,-3)

b. (3,3)

c. (-3,3)

d. **(3,-3)** ✓

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(3, -3)$.

43 [Challenging] • Target: Composite Transformations • Cross-Lesson: 8.01 Functions: Composite & Inverse

The curve $y = x^2$ is translated by the vector $(5; -3)$. State the coordinates of the new vertex.

a. **(5,-3)** ✓

b. (5,3)

c. (-5,-3)

d. (-3,5)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(5, -3)$.

44 [Challenging] • Target: Curve Sketching & Advanced Analysis • Cross-Lesson: 9.01 Circular Measure

The curve $y = x^2$ is translated by the vector $(-1; 5)$. State the coordinates of the new vertex.

a. (5,-1)

b. (1,5)

c. **(-1,5)** ✓

d. (-1,-5)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(-1, 5)$.

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45 [Challenging] • Target: Basic Graph Forms & Translations • Cross-Lesson: 1.01 Indices, Surds & Quadratics

The curve $y = x^2$ is translated by the vector $(1;\ -3)$. State the coordinates of the new vertex.

a. **(1,-3)** ✓

b. (-1,-3)

c. (-3,1)

d. (1,3)

A stretch parallel to the y-axis does not move the vertex $(0, 0)$; the translation then moves it to $(1,\ -3)$.

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